

Meeting Minutes: Determinants of Outsourcing Prices and Empirical Regression Design

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Contents

Note

These minutes were organized based on the content recorded during the voice discussion. Details that were not explicitly stated in the discussion are marked as items to be confirmed rather than added as assumptions.

1 Meeting Topic

The meeting focused on a research and modeling question:

How can the outsourcing-related theoretical model be simplified, and how can data be used to test the determinants of outsourcing prices?

The core idea discussed was that, at the current stage, the outsourcing price can be treated as an observable or constructible variable rather than being fully endogenized inside the theoretical model. The team can then use data to analyze how outsourcing prices are related to firm characteristics, market conditions, and input/output-related factors.

2 Main Discussion Points

2.1 Model Simplification

The discussion began with how to simplify the existing model. One important proposal was:

- treat the outsourcing price as an exogenous variable;
- avoid fully modeling how outsourcing prices are endogenously determined through firm-to-firm transactions, bargaining, or market equilibrium at this stage;
- first use data regressions to study the determinants of outsourcing prices;
- reduce the complexity of the theoretical model so that the current research can move forward more easily.

The advantage of this approach is that the model can initially focus on the main mechanism, while the formation of outsourcing prices can be examined empirically. If the empirical results are clear, the team can later decide whether the theoretical model should formally incorporate the price formation mechanism.

2.2 Empirical Objective

A key task discussed in the meeting was:

Use data to analyze which factors determine outsourcing prices.

In other words, the goal is not merely to take outsourcing prices as given, but also to understand their sources and variation empirically.

A basic empirical strategy would be to use the outsourcing price faced by a firm for a given product or input as the dependent variable and examine the following groups of explanatory variables:

1. firm-level characteristics;
2. product-level or market-level transaction conditions;
3. market information on purchase and sales prices;
4. firm fixed effects and other control variables;
5. productivity, input efficiency, and other production-related variables after production data are merged.

3 Dependent Variable: Definition of the Outsourcing Price

3.1 Basic Definition

The meeting clarified that the firm's purchase price can be used as a measure of the outsourcing price.

Specifically:

- the focus is on the price paid by the firm when purchasing a product or input from outside;
- it is not necessary to simultaneously consider whether the firm later sells the same product;
- regardless of inventory adjustment, the current stage focuses mainly on the purchase-side price;
- the outsourcing price can be interpreted as the unit cost faced by the firm when obtaining external products or inputs through the market or a transaction network.

Therefore, the outsourcing price in the empirical analysis can be initially defined as:

$$\text{Outsourcing Price}_{ipct} = \frac{\text{Purchase Value}_{ipct}}{\text{Purchase Quantity}_{ipct}}$$

where:

- i : firm;
- p : product;
- c : city or region;
- t : year or time period;
- Purchase Value: purchase value or invoice amount;
- Purchase Quantity: purchase quantity.

3.2 Constructing Prices Using Invoice Value Divided by Quantity

The discussion also mentioned that, if more detailed raw data can be obtained, unit prices can be constructed as:

$$\text{Unit Price} = \frac{\text{Invoice Value}}{\text{Quantity}}$$

That is, the unit price can be constructed by dividing invoice value by quantity.

This means that an important next step in the data work is to confirm whether sufficiently detailed raw transaction data are available, including both transaction value and quantity. If these variables are available, more precise transaction-level or firm-product-level price measures can be constructed.

4 Explanatory Variable Group 1: Firm Characteristics

The meeting discussed several firm-level explanatory variables that can be used to analyze which types of firms face higher or lower outsourcing prices.

4.1 Firm Size

Firm size is an important factor. The discussion noted that, before the data are fully merged, the following variables can be used as proxies for firm size:

- total sales;
- or other aggregate firm-level variables.

A preliminary specification could be:

$$\text{Outsourcing Price}_{ipct} = \beta_1 \text{Firm Size}_{it} + \dots$$

where firm size can temporarily be measured by total sales.

After more production data are merged, the team may also consider using:

- total output;
- total employment;
- total assets;
- total capital;
- total transaction scale;

as size measures or robustness checks.

4.2 Capital Size

The meeting also explicitly mentioned that capital size can be tested as a potential determinant.

Capital size may affect a firm's bargaining power, production capacity, outsourcing demand, and range of supplier choices. Therefore, the regression can include a capital variable such as:

$$\text{Capital}_{it}$$

This can be used to test whether firms with larger capital obtain lower outsourcing prices, or whether they pay higher prices because they purchase higher-quality products or inputs.

4.3 Production Quantity

Production quantity was also discussed as a possible explanatory variable.

Production quantity may affect purchase volume, purchase frequency, bargaining power, and supply-chain relationships. For example, firms with larger production quantities may:

- have more stable or larger-scale procurement needs;
- be more likely to build long-term relationships with suppliers;
- be more likely to obtain quantity discounts;
- also potentially purchase higher-quality outsourced products or inputs.

Therefore, production quantity can be an important firm characteristic in explaining outsourcing prices.

4.4 Productivity and Input Efficiency

The discussion also noted that productivity and input efficiency are important factors. However, before production data are merged, these indicators may not be directly available.

If production data can be successfully merged later, the following variables can be added:

- firm productivity;
- input efficiency;
- cost efficiency;
- other firm-level efficiency measures estimated from production functions.

These variables can help analyze:

- whether high-productivity firms obtain lower outsourcing prices;
- whether more efficient firms have stronger supplier selection ability;
- whether outsourcing prices reflect firm capability, bargaining power, or procurement quality.

5 Explanatory Variable Group 2: Market Conditions

In addition to firm characteristics, the meeting emphasized the importance of market conditions.

5.1 Product Market Transaction Activity

The discussion noted that the number of transactions for a given product in a market should be considered. That is, the level of transaction activity in a product market may affect outsourcing prices.

For example, the following variables can be constructed:

- the number of transactions for a product in a city;
- the transaction value for a product in a city;
- the number of participating firms for a product in a city;
- the number of buyers for a product in a city;
- the number of sellers for a product in a city.

These variables can reflect market thickness or market liquidity.

The basic logic is:

- if a product has many transactions in a city, the market is thicker;
- a thicker market may make it easier for firms to find suppliers;
- stronger competition may reduce purchase prices;
- however, if active transactions reflect strong demand, prices may also be higher.

Therefore, the direction of the effect of market transaction conditions on outsourcing prices needs to be tested empirically.

5.2 Average Transaction Price at the Product-City Level

The meeting explicitly mentioned calculating the average transaction price of each product within a city, regardless of whether the firm is a buyer or a seller.

This can serve as a market-level price condition that helps explain the outsourcing price faced by firms. For example:

$$\bar{P}_{pct}$$

represents the average transaction price of product p in city c at time t .

This variable can help control for the overall price level in the local product market, so that firm-level price differences are not fully conflated with differences in market-wide price levels.

5.3 Distinguishing Purchase Prices and Sales Prices

The meeting also discussed that average transaction prices can be further separated into:

- average purchase prices;
- average sales prices.

This distinction is important because purchase-side and sales-side prices may have different economic meanings:

- purchase prices are closer to the outsourcing cost faced by the firm as a buyer;
- sales prices are closer to the market price received by the firm as a supplier or seller;
- the gap between purchase and sales prices may reflect transaction costs, distribution margins, quality differences, or market power.

Therefore, the following variables can be constructed:

$$\bar{P}_{pct}^{buy}$$

$$\bar{P}_{pct}^{sell}$$

which represent the average purchase price and average sales price at the product-city-time level, respectively.

6 Demand-Side and Supply-Side Factors

The meeting noted that demand-side and supply-side factors should be distinguished.

Market conditions can be further decomposed into the following components.

6.1 Demand-Side Factors

Demand-side factors may include:

- total purchase quantity of a product within a city;
- number of buying firms;
- buyer concentration;
- strength of demand;
- downstream industry demand intensity;
- product transaction value or transaction frequency.

If demand is strong, the outsourcing price may rise because more firms compete to purchase the same product or input.

6.2 Supply-Side Factors

Supply-side factors may include:

- number of selling firms;
- seller concentration;
- total supply of a product within a city;
- degree of supplier competition;
- supplier production capacity or sales scale.

If supply is more abundant or supplier competition is stronger, purchase prices may fall. Conversely, if there are fewer suppliers or the market is more concentrated, outsourcing prices may be higher.

6.3 Market Prices Are Jointly Determined by Buyers and Sellers

The discussion can be summarized as follows: outsourcing prices are determined not only by the purchasing firm's own characteristics, but also by the interaction between buyers and sellers in the product market.

Therefore, the empirical analysis can include:

- firm-level characteristics;
- product-city-level demand conditions;
- product-city-level supply conditions;
- product-city average prices;
- time fixed effects;
- firm fixed effects;
- product or city fixed effects.

7 Fixed Effects and Control Variables

7.1 Firm Fixed Effects

The meeting specifically mentioned adding firm fixed effects.

Firm fixed effects help control for firm characteristics that do not vary over time but may affect outsourcing prices, such as:

- long-term firm productivity;
- managerial capability;
- long-term bargaining power;
- long-term supplier networks;
- quality positioning;
- stable differences in firm size or business model.

After adding firm fixed effects, identification relies more on variation within the same firm across different products, time periods, or markets.

A basic specification can be written as:

$$\log P_{ipct}^{buy} = \beta X_{it} + \gamma Z_{pct} + \alpha_i + \delta_t + \varepsilon_{ipct}$$

where:

- P_{ipct}^{buy} : the purchase price paid by firm i for product p in city c at time t ;
- X_{it} : firm-level variables;
- Z_{pct} : product-city-time-level market conditions;
- α_i : firm fixed effects;
- δ_t : time fixed effects;
- ε_{ipct} : error term.

7.2 Other Possible Fixed Effects

Depending on the data structure, the team can also consider adding:

- product fixed effects;
- city fixed effects;
- year fixed effects;
- product-year fixed effects;
- city-year fixed effects;
- product-city fixed effects.

These fixed effects can help control for unobserved heterogeneity at different levels.

8 Data Issues and Next Data Work

8.1 The I/O Table Does Not Contain Price Information

The meeting noted that the original input-output table does not contain price information.

This means that relying only on the I/O table may not allow the team to directly analyze outsourcing prices. Therefore, it is necessary to check whether more detailed raw transaction data can be obtained.

8.2 Full Data May Now Be Available

Later in the discussion, it was mentioned that the full data may now be available. This is important because, if the complete raw data can be obtained, the team can more accurately construct:

- transaction prices;
- purchase prices;
- sales prices;
- firm-product-city-level average prices;
- product-city market transaction conditions;
- firm purchase quantities and purchase values.

8.3 Confirm Whether Raw Data Include Value and Quantity

The next step is to confirm whether the data include:

- invoice value;
- transaction quantity;
- product code;
- firm code;
- buyer/seller identity;
- city or region information;
- time information.

If both value and quantity are available, unit prices can be constructed. If quantity is missing, the price variable will be substantially limited.

8.4 Random-Sample Testing

The meeting discussed first drawing a random sample of the data for testing.

The purpose of this step includes:

- checking data loading and processing speed;
- checking whether variables are complete;
- checking whether price variables can be constructed reasonably;
- understanding the difficulty of data cleaning;
- finding a balance between processing speed and reliability before processing the full large-scale data.

This step is a key preparation for the subsequent empirical work.

9 Preliminary Empirical Design

Based on the meeting discussion, the following preliminary empirical framework can be formed.

9.1 Stage 1: Construct Basic Variables

The team should first construct the following basic variables:

Type	Variable	Description
Dependent variable	Purchase unit price	Invoice value divided by quantity
Firm size	Total sales	Can be used before merging production data
Capital size	Capital variable	Tests the effect of capital size on price
Production scale	Output quantity	If data are available
Market thickness	Product-city transaction count	Measures market activity
Market price	Product-city average transaction price	Can distinguish purchase and sales prices
Demand conditions	Number of buyers, purchase quantity	Reflects demand strength
Supply conditions	Number of sellers, sales quantity	Reflects supply strength

9.2 Stage 2: Baseline Regression

The analysis can begin with a simple regression:

$$\log P_{ipct}^{buy} = \beta_1 \log Sales_{it} + \beta_2 \log Capital_{it} + \beta_3 MarketCondition_{pct} + \varepsilon_{ipct}$$

Then fixed effects can be added gradually:

$$\log P_{ipct}^{buy} = \beta X_{it} + \gamma Z_{pct} + \alpha_i + \delta_t + \varepsilon_{ipct}$$

9.3 Stage 3: Separate Demand-Side and Supply-Side Market Conditions

The specification can be further written as:

$$\log P_{ipct}^{buy} = \beta X_{it} + \gamma_1 Demand_{pct} + \gamma_2 Supply_{pct} + \gamma_3 \bar{P}_{pct} + FE + \varepsilon_{ipct}$$

where:

- $Demand_{pct}$: product-city-level demand conditions;
- $Supply_{pct}$: product-city-level supply conditions;
- $\bar{P}_{pct}$: average market transaction price;
- FE : the set of fixed effects.

9.4 Stage 4: Extension After Merging Production Data

If production data can later be merged, the following variables can be added:

- productivity;
- input efficiency;
- firm cost capability;
- product quality or quality proxy variables;
- firm production scale;
- product scope;
- other production-side variables.

This can further test whether outsourcing prices are related to firm capability, quality choice, or production efficiency.

10 Treatment of Purchase Prices, Sales Prices, and Inventory

The meeting discussed that the purchase price can currently be used as the outsourcing price, without considering whether the firm immediately sells the purchased product.

That is:

- the outsourcing price is defined from the purchase side;
- even if the firm adjusts inventory, this does not affect the current definition of the outsourcing price;
- sales quantity does not need to be used to redefine the purchase price;
- the focus is on the unit price paid by the firm as a buyer in the transaction.

This treatment helps keep the variable definition clear.

However, if future research focuses on inventory, resale, or carry-along trade, it may be necessary to further distinguish:

- purchases used for production;
- purchases used for resale;
- purchases that enter inventory;
- time gaps between purchases and sales.

These issues can be treated as extensions rather than the main task at the current stage.

11 Action Items

Based on the meeting discussion, the following action items can be summarized.

11.1 Data Acquisition and Confirmation

- Confirm whether the complete raw transaction data can be obtained.
- Confirm whether the data include invoice value and quantity.
- Confirm whether buyers, sellers, products, cities, and time can be identified.
- Confirm whether purchase transactions and sales transactions can be distinguished.

11.2 Data Sampling Test

- Randomly draw a subset of the data.
- Test data reading and processing speed.
- Check variable completeness.
- Construct unit prices using value divided by quantity.
- Check whether the price distribution is reasonable.
- Find a balance between processing speed and reliability.

11.3 Variable Construction

- Construct firm-level purchase prices.
- Construct product-city-level average transaction prices.
- Distinguish average purchase prices and average sales prices.
- Construct product-city transaction counts.
- Construct demand-side and supply-side market conditions.
- Construct firm size variables, such as total sales.
- After merging production data, construct productivity, input efficiency, and related variables.

11.4 Regression Analysis

- Begin with baseline regressions.
- Add firm size, capital, production quantity, and other variables.
- Add market condition variables.
- Add firm fixed effects.
- Gradually add product, city, year, and other fixed effects.
- Compare whether the results are robust across different specifications.

11.5 Follow-Up Communication

- If the data structure or variable definitions remain unclear, further confirmation with the data provider or collaborators is needed.
- It is especially important to confirm whether the raw data support the construction of transaction prices.
- If the data volume is very large, more efficient data processing approaches should be discussed.

12 Meeting Conclusions

The meeting produced a relatively clear research direction:

1. In the theoretical model, the outsourcing price can initially be treated as an exogenous variable in order to simplify the model.
2. In the empirical analysis, transaction data can be used to study the determinants of outsourcing prices.

3. The outsourcing price can initially be measured by the purchase unit price, namely invoice value divided by quantity.
4. Firm size, capital, production quantity, productivity, and input efficiency are potential explanatory variables.
5. Product-city market conditions are also important, including transaction counts, average transaction prices, demand-side factors, and supply-side factors.
6. Firm fixed effects can be used to control for long-term unobserved firm heterogeneity.
7. The key current task is to confirm whether complete raw data can be obtained and to first test data quality and processing speed using a random sample.
8. After the data are prepared, the team can gradually conduct regression analysis and decide, based on the results, whether the theoretical model needs to be extended.

13 Concise Action Checklist

- ☐ Confirm whether complete raw transaction data can be obtained.
- ☐ Check whether the raw data include value and quantity.
- ☐ Construct unit prices using value divided by quantity.
- ☐ Distinguish purchase prices and sales prices.
- ☐ Construct product-city-level average prices and transaction activity measures.
- ☐ Construct firm size, capital, and production quantity variables.
- ☐ First use total sales as a proxy for firm size.
- ☐ After merging production data, add productivity and input efficiency.
- ☐ First test data processing speed and reliability using a random sample.
- ☐ Run baseline regressions and gradually add fixed effects and market condition variables.